

1

D

Average car tyre radius, $R = 0.20$ m

Cross sectional radius $r = 0.10$ m

$$\text{Volume} = (2\pi R)(\pi r^2) = 0.039 \text{ m}^3$$

2

B

$$\text{Units on right side} = (\text{unit of } k)(A^2)(\text{m/m})^2 = (\text{unit of } k)(A^2)$$